Fraud Detection Pipeline

IMI Big Data & AI Competition · 6 Weeks

UofT & Scotiabank

Python · scikit-learn · pandas · NumPy · Matplotlib · Jupyter Notebook

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5.9M+
Transactions

61K+
Customers

7
Data Sources

0.895
AUC Score

The Problem

Banks lose billions annually to financial fraud. Manual review doesn't scale — analysts can't examine millions of transactions across multiple channels.

The challenge: build a machine learning system that automatically flags suspicious customers using transactional and demographic data from a real Canadian bank partner.

The complication: only 1% of customers are fraudulent — a severe class imbalance that causes simple models to just predict everyone as legitimate and call it 99% accurate.

5.9M+
Transactions analyzed

1,000
Labeled customers (for training)

~1%
Fraud rate (10 of 1,000)

Process

A 6-week sprint from stakeholder discovery to working prototype

Weeks 1–2

Discovery & Research

Workshop sessions with the bank's fraud team to understand detection challenges, analyst workflows, and pain points. Explored all 12 data files.

Weeks 3–4

Feature Engineering

Designed the feature pipeline — transforming raw transactions into customer-level behavioral profiles. Tested features based on insights from bank workshops.

Weeks 5–6

Modeling & Explainability

Trained and compared 3 models. Built the automated explanation system so flagged customers come with clear reasoning for analysts.

Irfan: data cleaning, EDA, explainability · Isha: data cleaning, EDA, documentation · Both: feature engineering, model training

Data & Approach

12 CSV files from the bank partner, spanning 7 transaction types + customer profiles

ATM	185K
Card	3.5M
Cheque	240K
EFT	1.07M
E-Transfer	845K
Wire / WU	7K

Pipeline

Load & Join — 12 CSV files → **Feature Eng.** — 67 features → **Train Models** — 3 algorithms → **Evaluate** — AUC + metrics → **Explain** — Auto-reports

Feature Engineering

Transforming 5.9M raw transactions into 67 predictive features per customer

Transaction Behavior

~56 features

Count, total, average, max, min, std deviation, debit/credit ratio — computed per channel (ATM, card, cheque, EFT, EMT, wire, WU)

Card-Specific Signals

~7 features

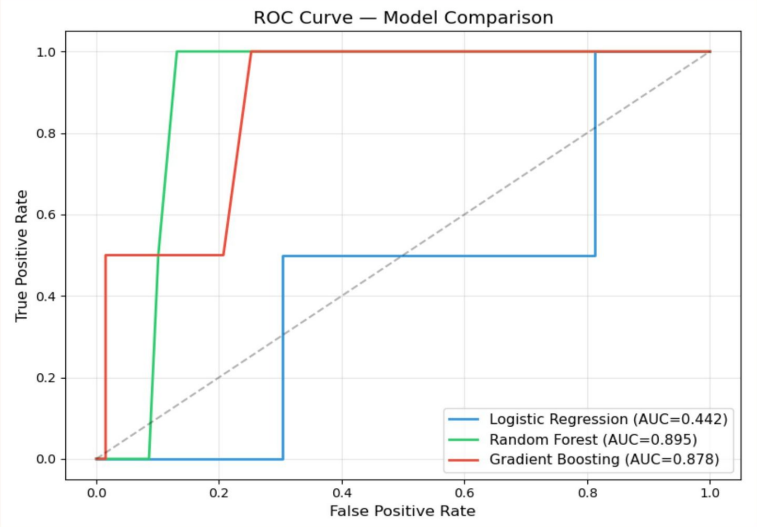
Ecommerce ratio, unique merchants, unique countries & cities, international transaction ratio

Customer Profile (KYC)

~6 features

Age, account age, income, gender, marital status, business flag, employee count

Model Performance



Model Comparison

Logistic Regression

AUC: 0.442 · Linear baseline — fails

Random Forest

AUC: 0.895 · Best performer

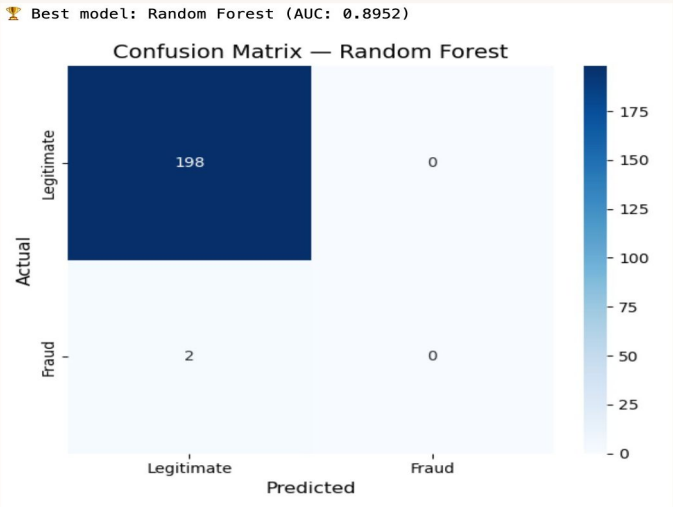
Gradient Boosting

AUC: 0.878 · Strong contender

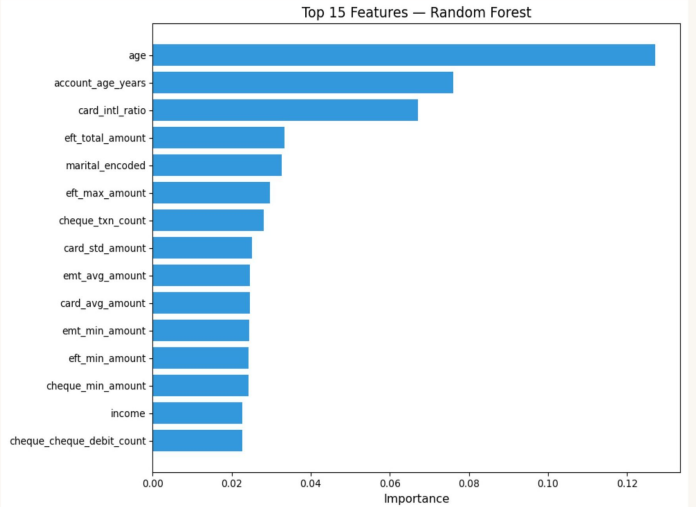
Non-linear models significantly outperform linear baselines — fraud patterns involve complex feature interactions

Detailed Results

Confusion Matrix



Feature Importance



AI-Powered Explainability

What an analyst would see: a ranked dashboard with automated risk reasoning

Fraud Risk Dashboard — Flagged Customers (sorted by risk score)

Customer ID	Risk Score	Status	Top Risk Indicators
SYNID0107832828	**0.94**	HIGH	Age: 18.6 · Account: 0.78 yrs · EMT avg: $534 · Card std: 6.82
SYNID0107464935	**0.91**	HIGH	Age: 17.3 · Account: 1.18 yrs · EFT total: $11,671 · EFT max: $2,611
SYNID0103892741	**0.72**	MEDIUM	Intl card ratio: 0.43 · Unique countries: 8 · Card std: 892.3
SYNID0105120384	**0.58**	MEDIUM	EFT total: $28,450 · Wire count: 14 · Account: 2.1 yrs
SYNID0102883956	**0.34**	LOW	Cheque count: 89 · ABM max: $2,000 · Income: $31K

The Deliverable

What the bank gets: an end-to-end scoring pipeline

Current State

Jupyter Notebook

Complete, reproducible pipeline. The bank provides their CSV files, the notebook processes them, engineers features, scores all 61K customers, and outputs a ranked list with explanations.

Next Step

Automated Script

Convert the notebook into a Python script that runs on a schedule — ingest new transaction data monthly, retrain if needed, and output updated risk scores automatically.

Production Vision

Cloud API + Dashboard

Deploy the model as an API endpoint on a cloud platform. Real-time scoring as transactions come in. Analyst-facing dashboard with drill-down views and exportable reports.

Learnings & Next Steps

What We Learned

- Feature engineering matters more than model choice — domain knowledge is the differentiator
- Class imbalance needs deliberate handling — AUC over accuracy, balanced class weights
- Explainability is essential — stakeholders need to trust and verify model decisions
- Real data is messy — missing values, mixed types, cross-channel complexity require careful cleaning

What We'd Improve

- Temporal features — transaction velocity, time-of-day patterns, behavioral drift over time
- LLM integration — natural language case narratives via Claude API for analyst reports
- Cloud deployment — real-time scoring API with an analyst-facing dashboard
- User testing — validate outputs with actual fraud analysts to refine what's most useful

Thank you — happy to discuss any of the technical details or our approach.